



**KNOX GRAMMAR SCHOOL**  
**MATHEMATICS DEPARTMENT**

**2004**  
**TRIAL HSC EXAMINATION**

# Mathematics

## Extension 2

### General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

### Total marks (120)

- Attempt Questions 1-8
- All questions are of equal value
- Use a **SEPARATE** Writing Booklet for each question
- Write your **Student Number** and **Teacher's Initials** on the front cover of each writing booklet

NAME: \_\_\_\_\_

TEACHER: \_\_\_\_\_

**Total marks (120)**

**Attempt questions 1 – 8**

**All questions are of equal value**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

| Question 1 (15 marks)                                                                                | Use a SEPARATE writing booklet. | Marks |
|------------------------------------------------------------------------------------------------------|---------------------------------|-------|
| (a) Evaluate $\int_0^{1.5} \frac{2}{\sqrt{9-x^2}} dx$ .                                              |                                 | 2     |
| (b) Find $\int \frac{1}{\sqrt{x^2-4x+5}} dx$ , with the aid of the Table of Standard Integrals.      |                                 | 2     |
| (c) Find $\int \sin^2 x \cos^3 x dx$ .                                                               |                                 | 3     |
| (d) Using the substitution $x = 3\sec\theta$ , evaluate $\int_3^6 \frac{1}{x^2\sqrt{x^2-9}} dx$ .    |                                 | 4     |
| (e) (i) Find constants $A, B, C$ such that $\frac{x^2+2}{x^2-x-2} \equiv A + \frac{Bx+C}{x^2-x-2}$ . |                                 | 1     |
| (ii) Hence find $\int \frac{x^2+2}{x^2-x-2} dx$ .                                                    |                                 | 3     |

**Question 2** (15 marks) Use a SEPARATE writing booklet.

**Marks**

(a) Simplify  $\frac{1-i^3}{1-i}$ . 2

(b) Let  $z = \frac{8-i}{2+i}$ .

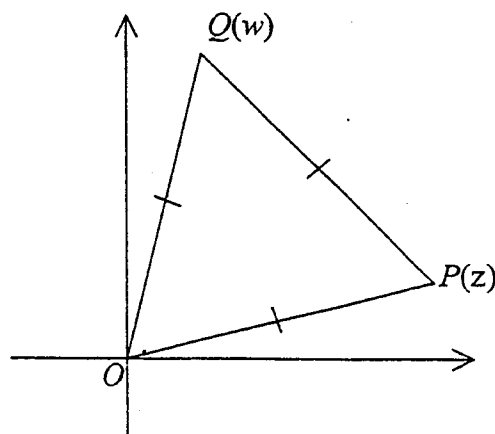
(i) Express  $z$  in the form  $a+bi$  where  $a$  and  $b$  are real numbers. 2

(ii) Hence, or otherwise, find  $|z|$  and  $\arg z$  (to 3 significant figures in the domain  $-\pi < \theta \leq \pi$ ). 3

(c) Sketch the region in the complex number plane where the inequalities  $|z+1-2i| \leq 2$  and  $\operatorname{Re}(z) \leq 0$  hold simultaneously. 2

(d) Factorise  $x^4 + 7x^2 - 18$  into the product of linear factors over the complex field. 2

(e)



In the Argand diagram,  $OPQ$  is an equilateral triangle.  $P$  represents the complex number  $z$  and  $Q$  represents the complex number  $w$ .

(i) Explain why  $w = z \operatorname{cis} \frac{\pi}{3}$ . 2

(ii) Show that  $w^3 + z^3 = 0$ . 2

- (a) Let  $f(x) = 2(x-1)(x-3)$ .

Draw separate sketches of the following functions (at least one-third of a page), showing clearly the important features, including any intercepts on the axes, turning points, asymptotes, etc.

- |       |                      |   |
|-------|----------------------|---|
| (i)   | $y = f(x)$           | 1 |
| (ii)  | $y = \frac{1}{f(x)}$ | 2 |
| (iii) | $y = 2 - f(x)$       | 2 |
| (iv)  | $y = \sqrt{f(x)}$    | 2 |
| (v)   | $y = \log_e f(x)$    | 2 |

- (b) Let  $I_n = \int_0^1 x^n e^{-x} dx$ .

- |       |                                                             |   |
|-------|-------------------------------------------------------------|---|
| (i)   | Evaluate $I_0$ .                                            | 1 |
| (ii)  | Prove that $I_n = n I_{n-1} - \frac{1}{e}$ for $n \geq 1$ . | 3 |
| (iii) | Hence evaluate $\int_0^1 x^3 e^{-x} dx$ .                   | 2 |

**Question 4** (15 marks) Use a SEPARATE writing booklet.

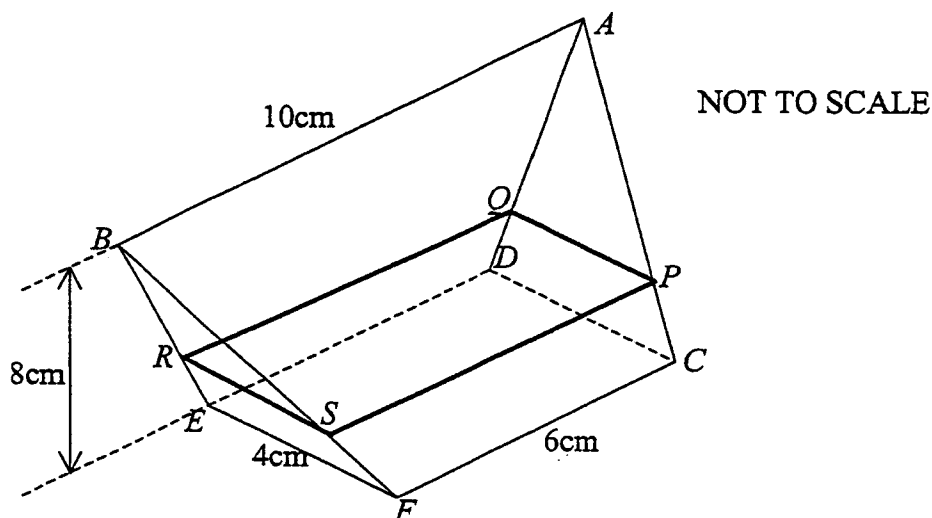
**Marks**

- (a) Find  $\sqrt{9-12i}$ . 3
- (b)  $(2+i)$  is a zero of the polynomial  $P(z) = z^3 - z^2 + az + b$ , where  $a$  and  $b$  are real numbers. 4
- Find the other two zeros, and the values of  $a$  and  $b$ .
- (c)  $\alpha, \beta, \gamma$  are the roots of the equation  $x^3 - 6x^2 + 12x - 35 = 0$ .
- (i) Form a cubic equation whose roots are  $\alpha - 2, \beta - 2, \gamma - 2$ . 2
- (ii) Hence, or otherwise, solve the equation  $x^3 - 6x^2 + 12x - 35 = 0$  over the complex field. 2
- (d) The roots of the equation  $z^2 + 5z - 2i = 0$  are  $\alpha$  and  $\beta$ . Without solving this equation, form the cubic equation whose roots are  $\alpha, \beta$  and  $(\alpha + \beta)$ . 4

(a) Consider the hyperbola  $\frac{x^2}{4} - \frac{y^2}{16} = 1$ .

- (i) Find its eccentricity. 1
- (ii) State the equations of the asymptotes. 1

(b)



The diagram shows a wedge with the edge  $AB$  parallel to the horizontal rectangular base  $CDEF$ , and the plane  $ABED$  is vertical.  $AB$  is 8 cm vertically above  $DE$ .  $PQRS$  is a rectangular cross-section  $h$  cm above the base.

- (i) Show that the area of the cross-section  $PQRS$  is  $\left(6 + \frac{h}{2}\right)\left(4 - \frac{h}{2}\right) \text{ cm}^2$ . 2
- (ii) Hence find the volume of the wedge. 2

(c) Consider the function  $y = \frac{x^2 - 3x}{x + 1}$ .

- (i) Find the equations of the two asymptotes. 2
- (ii) Find the coordinates of the stationary points and determine their nature. 5
- (iii) Sketch the graph of the function. 1
- (iv) For what values of  $k$  does the equation  $\frac{x^2 - 3x}{x + 1} = k$  have two real roots? 1

**Question 6** (15 marks) Use a SEPARATE writing booklet.

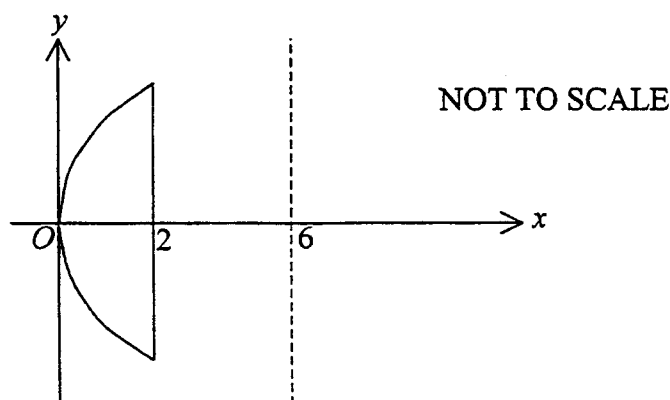
**Marks**

(a) (i) Show that  $f(x) = x\sqrt{4-x^2}$  is an odd function. 1

(ii) Hence, without finding any primitives, evaluate 2

$$\int_{-2}^2 (x\sqrt{4-x^2} - \sqrt{4-x^2}) dx, \text{ giving reasons.}$$

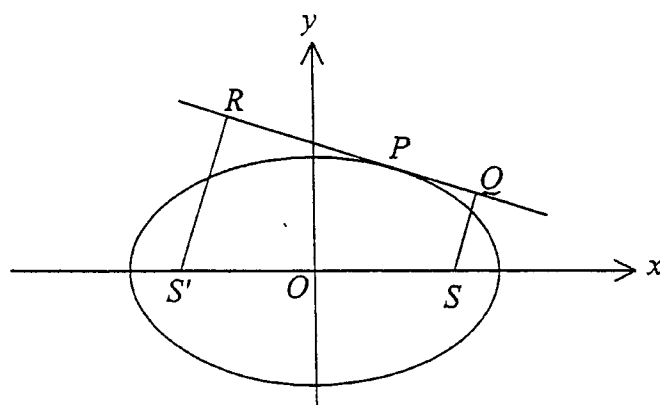
(b)



The region bounded by the parabola  $y^2 = 4x$  and the line  $x = 2$  is rotated about the line  $x = 6$ . 5

Using the method of cylindrical shells, find the volume of the solid formed.

(c)



(i) Prove that the equation of the tangent to the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  at the point  $P(a \cos \theta, b \sin \theta)$  is  $(b \cos \theta)x + (a \sin \theta)y - ab = 0$ . 3

(ii)  $Q$  and  $R$  are the feet of the perpendiculars to the tangent from the foci  $S$  and  $S'$  respectively. 4

Prove that  $SQ \times S'R = b^2$ .

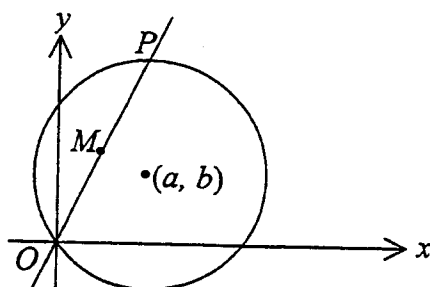
**Question 7** (15 marks) Use a SEPARATE writing booklet.

**Marks**

- (a) Find the general solution of the inequality  $\cos \theta \geq \frac{1}{2}$ .

2

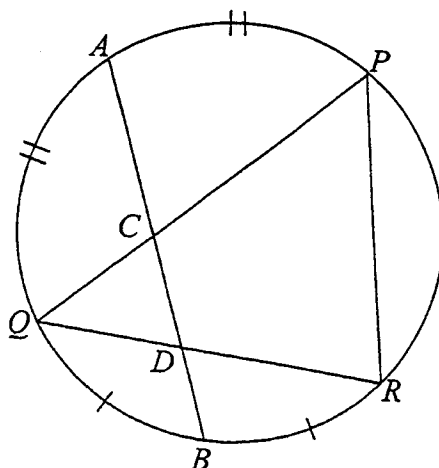
(b)



The diagram shows the graph of the circle  $(x-a)^2 + (y-b)^2 = a^2 + b^2$ , which passes through the origin  $O$ . The line  $y = mx$  cuts the circle at  $O$  and  $P$ .

- (i) Show that the  $x$  coordinate of  $P$  is  $\frac{2(a+bm)}{1+m^2}$ . 2
- (ii) Hence write down the coordinates of  $M$ , the midpoint of  $OP$ . 2
- (iii) Hence show that the locus of  $M$ , as the gradient of  $OP$  varies, is a circle, and state its centre and radius. 4

(c)



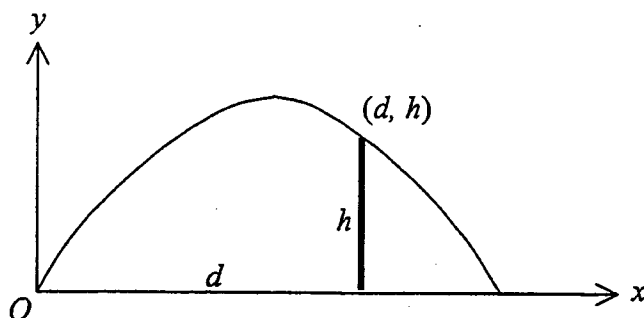
A circle is drawn through the vertices of the triangle  $PQR$ .  $A$  is the midpoint of the arc  $PQ$  and  $B$  is the midpoint of the arc  $QR$ . The chord  $AB$  intersects  $PQ$  at  $C$  and  $QR$  at  $D$ .

Copy or trace the diagram into your Writing Booklet.

- (i) Explain why  $\angle QPB = \angle BPR$ . 1
- (ii) Prove that  $QC = QD$ . 4



(a)



A stone is projected from a point on the ground, and it just clears a fence  $d$  metres away. The height of the fence is  $h$  metres. The angle of projection to the horizontal is  $\theta$  and the speed of projection is  $V$  m/s.

The displacement equations, measured from the point of projection, are:

$$x = V \cos \theta t \quad \text{and} \quad y = V \sin \theta t - \frac{1}{2} g t^2.$$

(i) Show that  $V^2 = \frac{gd^2 \sec^2 \theta}{2(d \tan \theta - h)}$ . 2

(ii) Show that the maximum height reached is  $\frac{d^2 \tan^2 \theta}{4(d \tan \theta - h)}$ . 3

(iii) Show that the stone will just clear the fence at its highest point if  $\tan \theta = \frac{2h}{d}$ . 3

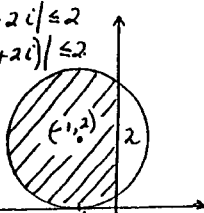
(b) (i) Prove by mathematical induction that  $(\sqrt{3} - 1)^n = p_n + q_n \sqrt{3}$ , where  $n$  is a positive integer and  $p_n$  and  $q_n$  are unique integers. 5

(ii) Hence show that  $p_n^2 - 3q_n^2 = (-2)^n$ . 2

End of paper

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                               | Comments                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <u>QUESTION 1</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                           |
| $(a) \int_0^{1.5} \frac{2}{\sqrt{9-x^2}} = 2 \left[ \sin^{-1} \frac{x}{3} \right]_0^{1.5} \quad \checkmark$ $= 2 \left[ \sin^{-1} \frac{1}{2} - \sin^{-1} 0 \right]$ $= 2 \left[ \frac{\pi}{6} - 0 \right]$ $= \frac{\pi}{3} \quad \checkmark \quad (2)$                                                                                                                                                                                                             |                                                                                                                           |
| $(b) \int \frac{1}{\sqrt{x^2-4x+5}} dx = \int \frac{1}{\sqrt{(x-2)^2+1}} dx \quad \checkmark$ $= \log_e(x-2 + \sqrt{(x-2)^2+1}) + C$ $\text{or } \log_e(x-2 + \sqrt{x^2-4x+5}) + C \quad (2)$                                                                                                                                                                                                                                                                        | Ignore 'c' in marking.                                                                                                    |
| $(c) \int \sin^2 x \cos^3 x dx$ $= \int \sin^2 x (1 - \sin^2 x) \cos x dx$ $= \int u^2 (1 - u^2) du \quad \checkmark$ $= \int (u^2 - u^4) du$ $= \frac{1}{3} u^3 - \frac{1}{5} u^5 + C \quad \checkmark$ $= \frac{1}{3} \sin^3 x - \frac{1}{5} \sin^5 x + C \quad \checkmark \quad (3)$                                                                                                                                                                              | $u = \sin x$<br>$du = \cos x dx$<br>or equivalent without substitution                                                    |
| $(d) \int_3^6 \frac{1}{x^2 \sqrt{x^2-9}} dx$ $= \int_0^{\frac{\pi}{3}} \frac{3 \sec \theta \tan \theta d\theta}{9 \sec^2 \theta \sqrt{9 \sec^2 \theta - 9}} \quad \checkmark$ $= \int_0^{\frac{\pi}{3}} \frac{3 \sec \theta \tan \theta d\theta}{9 \sec^2 \theta \times 3 \tan \theta}$ $= \frac{1}{9} \int_0^{\frac{\pi}{3}} \cos \theta d\theta \quad \checkmark$ $= \frac{1}{9} [\sin \theta]_0^{\frac{\pi}{3}} = \frac{\sqrt{3}}{18} \quad \checkmark \quad (4)$ | $x = 3 \sec \theta$<br>$dx = 3 \sec \theta \tan \theta d\theta$<br>$x = 3, \theta = 0$<br>$x = 6, \theta = \frac{\pi}{3}$ |

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                   | Comments                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <u>QUESTION 1 (cont.)</u>                                                                                                                                                                                                                                                                                                |                                                                                                                   |
| $(e) (i) \frac{1}{x^2-x-2} = \frac{1}{x^2+0x+2}$ $\frac{x^2-x-2}{x^2-x-2} = \frac{x^2+0x+2}{x^2-x-2}$ $\frac{x^2+2}{x^2-x-2} = 1 + \frac{x+4}{x^2-x-2}$ $A=1, B=1, C=4 \quad (1)$                                                                                                                                        | <u>OR</u><br>$\frac{x^2+2}{x^2-x-2} = A + \frac{Bx+C}{x^2-x-2}$ $x^2+2 = A(x^2-x-2) + Bx+C$ - equate coefficients |
| $(ii) \frac{x+4}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1}$ $x+4 = A(x+1) + B(x-2)$ $\text{Sub. } x=-1: 3 = -3B$ $\therefore B = -1$ $\text{Sub. } x=2: 6 = 3A$ $A = 2$ $\therefore \int \frac{x^2+2}{x^2-x-2} = \int \left( 1 + \frac{2}{x-2} - \frac{1}{x+1} \right) dx$ $= x + 2 \ln(x-2) - \ln(x+1) + C \quad (3)$ | 1 mark - correct A, B.<br><br>2 marks for 3 correct primitives                                                    |

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Comments |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <p><u>QUESTION 2</u></p> <p>a) <math>\frac{1-i}{1-i} = \frac{1+i}{1-i} \times \frac{1+i}{1+i}</math> OR <math>\frac{(1-i)(1+i+i^2)}{1-i}</math> ✓</p> $= \frac{1+2i+i^2}{1-i^2} = \frac{1+i-1}{1-(-1)} = \frac{i}{2}$ <p>②</p> <hr/> <p>(b) <math>z = \frac{8-i}{2+i}</math></p> <p>(i) <math>z = \frac{8-i}{2+i} \times \frac{2-i}{2-i}</math></p> $= \frac{16-10i+i^2}{4-i^2} = \frac{15-10i}{5} = 3-2i$ <p>②</p> <p>(ii) <math> z  = \sqrt{3^2+(-2)^2} = \sqrt{13}</math> ✓</p> $\tan \theta = -\frac{2}{3}$ ✓ $\theta = -0.588$ ✓ <p>③</p> <hr/> <p>(c) <math> z+1-2i  \leq 2</math><br/> <math> z-(-1+2i)  \leq 2</math></p>  <p><math>\text{Re}(z) \leq 0</math><br/> <math>x \leq 0</math></p> <p>1 - circle correct<br/> 1 - shading correct.</p> <p>②</p> |          |

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Comments                                                                                                                                    |
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| <p><u>QUESTION 2 (cont)</u></p> <p>(d) <math>x^4 + 7x^2 - 18</math></p> $= (x^2+9)(x^2-2)$ $= (x^2-9i^2)(x^2-2)$ $= (x-3i)(x+3i)(x-\sqrt{2})(x+\sqrt{2})$ <p>②</p> <hr/> <p>(e)</p> <p>(i) Since <math>OP = OQ</math> and <math>\angle POQ = \frac{\pi}{3}</math>, multiplying <math>z</math> by <math>\text{cis } \frac{\pi}{3}</math> doesn't change the modulus, as <math> \text{cis } \frac{\pi}{3}  = 1</math>, but adds <math>\frac{\pi}{3}</math> to the argument.</p> <p>②</p> <hr/> <p>(ii) <math>w = z \text{cis } \frac{\pi}{3}</math></p> $w^3 = (z \text{cis } \frac{\pi}{3})^3$ $= z^3 \text{cis } \pi$ $= z^3 (\cos \pi + i \sin \pi)$ $= z^3 (-1 + 0i)$ $= -z^3$ <p><math>\therefore w^3 + z^3 = 0</math> ✓</p> <p>②</p> | <p>1 - for factors of <math>x^2+9</math><br/> 1 - for factors of <math>x^2-2</math>.</p> <p>1 - modulus reason<br/> 1 - argument reason</p> |

| Suggested Solution (s)                                                                                                                                                                                                                       | Comments |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <p><u>QUESTION 3.</u></p> <p>a) <math>f(x) = 2(x-1)(x-3)</math></p> <p>(i)  <math>y = f(x)</math> ①</p> <p>(ii)  <math>y = \frac{1}{f(x)}</math> ②</p> <p>(iii)  <math>y = 2 - f(x)</math> ②</p> <p>(iv)  <math>y = \sqrt{f(x)}</math> ②</p> |          |

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Comments |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <p><u>QUESTION 3 (cont.)</u></p> <p>a)(v)  <math>y = \log_e f(x)</math> ②</p> <p>(b) <math>I_n = \int_0^1 x^n e^{-x} dx</math></p> <p>(i) <math>I_0 = \int_0^1 e^{-x} dx</math><br/> <math>= [-e^{-x}]_0^1</math><br/> <math>= (-e^{-1}) - (-e^0)</math><br/> <math>= 1 - \frac{1}{e}</math> ①</p> <p>(ii) <math>I_n = \int_0^1 x^n \frac{d}{dx}(e^{-x}) dx</math><br/> <math>= [-x^n e^{-x}]_0^1 - \int_0^1 (-e^{-x}) n x^{n-1} dx \checkmark</math><br/> <math>= [-e^{-1} - 0] + n \int_0^1 x^{n-1} e^{-x} dx \checkmark</math><br/> <math>= -\frac{1}{e} + n I_{n-1}</math> ③</p> <p>(iii) <math>\int_0^1 x^3 e^{-x} dx = I_3</math><br/> <math>= 3 I_2 - \frac{1}{e}</math><br/> <math>= 3 [2 I_1 - \frac{1}{e}] - \frac{1}{e} \checkmark</math><br/> <math>= 6 I_1 - \frac{4}{e}</math><br/> <math>= 6 [1 \times I_0 - \frac{1}{e}] - \frac{4}{e}</math><br/> <math>= 6 I_0 - \frac{10}{e}</math><br/> <math>= 6 (1 - \frac{1}{e}) - \frac{10}{e}</math><br/> <math>= 6 - \frac{16}{e} \checkmark</math> ②</p> |          |

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Comments                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <p><u>QUESTION 4.</u></p> <p>a) Let <math>9-12i = x+yi</math> (<math>x, y</math> real)</p> $9-12i = x^2-y^2+2xyi$ $x^2-y^2=9, \quad 2xy=-12 \quad \checkmark$ $x^2-\frac{36}{x^2}=9 \quad y=-\frac{6}{x}$ $x^4-9x^2-36=0$ $(x^2-12)(x^2+3)=0$ $x=\pm 2\sqrt{3} \quad (x \text{ real}) \quad \checkmark$ <p>If <math>x=2\sqrt{3}, y=-\frac{6}{2\sqrt{3}}=-\sqrt{3}</math></p> <p>If <math>x=-2\sqrt{3}, y=-\frac{6}{-2\sqrt{3}}=\sqrt{3}</math></p> $\therefore \sqrt{9-12i} = \pm(2\sqrt{3}-\sqrt{3}i) \quad \checkmark \quad (3)$ <p>b) <math>P(z) = z^3 - z^2 + az + b</math>.</p> <p><math>(2+i)</math> is a root <math>\therefore (2-i)</math> is a root <math>\checkmark</math></p> <p>By sum of roots: <math>(2+i)+(2-i)+\alpha = 1</math></p> $4+\alpha = 1$ $\alpha = -3 \quad \checkmark$ <p>Roots are <math>2+i, 2-i, -3</math>.</p> <p>Factors are <math>(z+3)(z^2-4z+5)</math></p> $= z^3 - z^2 - 7z + 15$ $\therefore a = -7, b = 15 \quad \checkmark \quad (4)$ <p>c) <math>x^3 - 6x^2 + 12x - 35 = 0</math>.</p> <p>i) Roots <math>\alpha-2, \beta-2, \gamma-2</math>.</p> <p>Let <math>y = x-2</math> i.e. <math>x = y+2</math></p> $(y+2)^3 - 6(y+2)^2 + 12(y+2) - 35 = 0 \quad \checkmark$ $y^3 + 6y^2 + 12y + 8 - 6y^2 - 24y - 24 + 12y + 24 - 35 = 0$ $y^3 - 27 = 0 \quad \checkmark \quad (2)$ | <p>Other methods could be:</p> <p>Subst. <math>z = 2+i</math></p> <p>equate real &amp; imag. parts, etc.</p> |

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Comments                                                                                                                                                                                                                                                                                                                                                                                            |
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| <p><u>QUESTION 4 (cont.)</u></p> <p>(c) (ii) <math>(y-3)(y^2+3y+9) = 0</math></p> $y = 3 \text{ or } \frac{-3 \pm \sqrt{9-36}}{2} \quad \checkmark$ $y = 3 \text{ or } y = \frac{-3+3\sqrt{3}i}{2} \text{ or } y = \frac{-3-3\sqrt{3}i}{2}$ $x = 5, \frac{1}{2} + \frac{3\sqrt{3}}{2}i, \frac{1}{2} - \frac{3\sqrt{3}}{2}i \quad \checkmark \quad (2)$ <p>(d) <math>z^2 + 5z - 2i = 0</math>.</p> <p>Roots <math>\alpha, \beta, \alpha+\beta</math>.</p> $\Sigma \alpha: \alpha + \beta + (\alpha + \beta) = 2(\alpha + \beta)$ $= 2 \times (-5)$ $= -10 \quad \checkmark$ $\Sigma \alpha\beta: \alpha\beta + \alpha(\alpha+\beta) + \beta(\alpha+\beta)$ $= 3\alpha\beta + \alpha^2 + \beta^2$ $= (\alpha+\beta)^2 + \alpha\beta$ $= (-5)^2 + (-2i)$ $= 25 - 2i \quad \checkmark$ $\alpha\beta\gamma: \alpha\beta(\alpha+\beta) = (-2i)(-5)$ $= 10i \quad \checkmark$ <p><math>\therefore</math> Cubic equation with roots <math>\alpha, \beta, \alpha+\beta</math>:</p> $z^3 + 10z^2 + (25-2i)z - 10i = 0 \quad \checkmark \quad (4)$ | <p>OR</p> <p>1-for <math>x=5</math> by any method (eg. factor theorem)</p> <p>1-other values correct</p> <p><math>\rightarrow</math> or equivalent</p> <p><math>\rightarrow</math> or equivalent.</p> <p>OR</p> <p><math>\alpha + \beta = -5</math></p> <p>Equation is <math>(z+5)(z^2+5z-2i)=0</math></p> $z^3 + 5z^2 - 2iz + 5z^2 + 25z - 10i = 0$ <p>i.e.</p> $z^3 + 10z^2 + (25-2i)z - 10i = 0$ |

Suggested Solution (s)

Comments

QUESTION 5.

a)  $\frac{x^2}{4} - \frac{y^2}{16} = 1$

$a^2 = 4$

$b^2 = 16$

(i)  $b^2 = a^2(e^2 - 1)$

$a = 2$

$16 = 4(e^2 - 1)$

$b = 4$

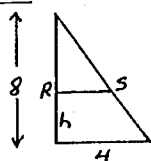
$e = \sqrt{5}$

(ii) Asymptotes:  $y = \pm \frac{b}{a}x$

$y = \pm 2x$

b)(i) End view:  $\frac{RS}{4} = \frac{8-h}{8}$

$RS = \frac{4}{8}(8-h)$   
 $= 4 - \frac{1}{2}h$



Back surface:

$\frac{x}{2} = \frac{h}{8}$   
 $x = \frac{h}{4}$

$RQ = 6 + 2 \times \frac{h}{4}$   
 $= 6 + \frac{h}{2}$

$\therefore \text{Area PQRS} = (6 + \frac{h}{2})(4 - \frac{1}{2}h)$

(ii)  $V = \int_0^8 (6 + \frac{h}{2})(4 - \frac{1}{2}h) dh$   
 $= \int_0^8 (24 - h - \frac{h^2}{4}) dh$   
 $= [24h - \frac{1}{2}h^2 - \frac{1}{12}h^3]_0^8$   
 $= (24 \times 8 - \frac{1}{2} \times 64 - \frac{1}{12} \times 512)$

Volume =  $117 \frac{1}{3} \text{ cm}^3$

Suggested Solution (s)

Comments

QUESTION 5 (cont.)

(c)  $y = \frac{x^2 - 3x}{x+1}$

(i)  $x+1 \overline{) x^2 - 3x + 0}$   
 $\underline{x^2 + x}$   
 $-4x + 0$   
 $\underline{-4x - 4}$   
 $4$

$y = x - 4 + \frac{4}{x+1}$

Asymptotes:  $x = -1$ ,  $y = x - 4$

(ii)  $\frac{dy}{dx} = \frac{(x+1)(2x-3) - (x^2-3x)(1)}{(x+1)^2}$

$= \frac{2x^2 + 2x - 3}{(x+1)^2}$

$= \frac{(x+3)(x-1)}{(x+1)^2}$

stat. pts. at  $x = -3$ ,  $x = 1$

Stat. pts. at  $(-3, -9)$ ,  $(1, -1)$

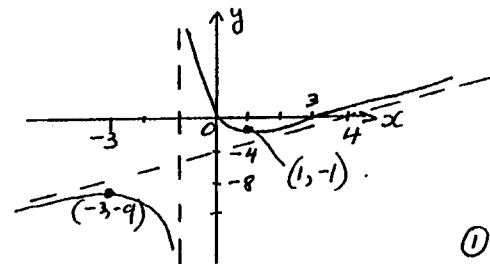
$f'(-3-\epsilon) = \frac{(-)(-)}{(+)} > 0$ ;  $f'(-3+\epsilon) = \frac{(+)(-)}{(+)} < 0$

$\therefore$  Rel. maximum at  $(-3, -9)$

$f'(1-\epsilon) = \frac{(+)(-)}{(+)} < 0$ ;  $f'(1+\epsilon) = \frac{(+)(+)}{(+)} > 0$

$\therefore$  Rel. minimum at  $(1, -1)$

(iii).



(iv). Two real roots for  $k > -1$ ,  $k < -9$

## Suggested Solution (s)

## Comments

QUESTION 6

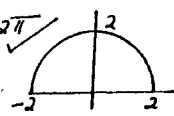
2) (i)  $f(x) = x\sqrt{4-x^2}$

$$f(-x) = (-x)\sqrt{4-(-x)^2} = -x\sqrt{4-x^2} = -f(x)$$

 $\therefore f(x)$  is odd function (1)

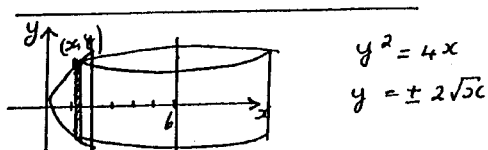
(ii)  $\int_{-2}^2 x\sqrt{4-x^2} dx = 0$  ✓ since  $f(x)$  is odd.

$$\int_{-2}^2 \sqrt{4-x^2} dx = \frac{1}{2} \times \pi \times 2^2 = 2\pi$$
  
since it represents area of a semi-circle



$$\therefore \int_{-2}^2 (x\sqrt{4-x^2} - \sqrt{4-x^2}) dx = 0 - 2\pi = -2\pi$$
 (2)

b).



$$y^2 = 4x$$
  
$$y = \pm 2\sqrt{x}$$

Radius =  $6 - x$  ✓

$$\delta V \doteq 2\pi(6-x)2y \delta x$$

$$\doteq 4\pi(6-x)y \delta x$$
 ✓

$$V = 4\pi \int_0^2 (6-x)y dx$$

$$= 4\pi \int_0^2 (6-x)2\sqrt{x} dx$$
 ✓

$$= 4\pi \int_0^2 (12x^{\frac{1}{2}} - 2x^{\frac{3}{2}}) dx$$

$$= 4\pi \left[ 12 \times \frac{2}{3} x^{\frac{3}{2}} - 2 \times \frac{2}{5} x^{\frac{5}{2}} \right]_0^2$$
 ✓

$$= 4\pi \left[ 8 \times 2\sqrt{2} - \frac{4}{5} \times 2^{\frac{5}{2}} \right]$$

$$= 4\pi \left[ \frac{80\sqrt{2}}{5} - \frac{16\sqrt{2}}{5} \right]$$

$$\text{Volume} = \frac{256\sqrt{2}\pi}{5} \text{ unit}^3$$
 ✓

## Suggested Solution (s)

## Comments

QUESTION 6 (cont.)

(c)  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(i)  $\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$
 ✓

At  $P(a\cos\theta, b\sin\theta)$ ,  $\frac{dy}{dx} = -\frac{b^2 a \cos\theta}{a^2 b \sin\theta}$   
 $= -\frac{b \cos\theta}{a \sin\theta}$

Tangent:  $y - b\sin\theta = -\frac{b \cos\theta}{a \sin\theta} (x - a\cos\theta)$  ✓

$$(a \sin\theta)y - ab \sin^2\theta = -(b \cos\theta)x + ab \cos^2\theta$$

$$(b \cos\theta)x + (a \sin\theta)y = ab(\cos^2\theta + \sin^2\theta)$$

$$(b \cos\theta)x + (a \sin\theta)y - ab = 0$$
 ✓

(3)

(ii)  $S(ae, 0) \quad S'(-ae, 0)$

$$SQ \times S'R = \left| \frac{(b \cos\theta)(ae) - ab}{\sqrt{b^2 \cos^2\theta + a^2 \sin^2\theta}} \right| \times \left| \frac{(b \cos\theta)(-ae) - ab}{\sqrt{b^2 \cos^2\theta + a^2 \sin^2\theta}} \right|$$
 ✓

$$= \left| \frac{ab(e \cos\theta - 1)(-ae \cos\theta - 1)}{b^2 \cos^2\theta + a^2 \sin^2\theta} \right|$$

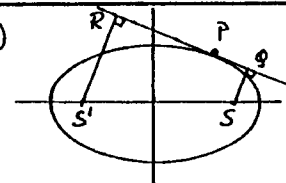
$$= \left| \frac{a^2 b^2 (e^2 \cos^2\theta - 1)}{a^2 (1 - e^2) \cos^2\theta + a^2 \sin^2\theta} \right|$$
 ✓

$$= \left| \frac{b^2 (e^2 \cos^2\theta - 1)}{\cos^2\theta - e^2 \cos^2\theta + \sin^2\theta} \right|$$
 ✓

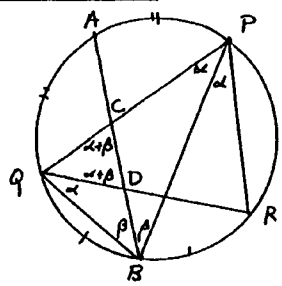
$$= \left| \frac{b^2 (e^2 \cos^2\theta - 1)}{1 - e^2 \cos^2\theta} \right|$$

$$= b^2$$
 ✓

(4)



| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Comments                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <p><u>QUESTION 7.</u></p> <p>a) <math>\cos \theta \geq \frac{1}{2}</math><br/> <math>2n\pi - \frac{\pi}{3} \leq \theta \leq 2n\pi + \frac{\pi}{3}</math> (2)</p> <p>b) <math>y = mx</math>; <math>(x-a)^2 + (y-b)^2 = a^2 + b^2</math><br/>         (i) <math>x^2 + y^2 - 2ax - 2by = 0</math><br/>         Sub. <math>y = mx</math>: <math>x^2 + m^2x^2 - 2ax - 2bmx = 0</math><br/> <math>x^2(1+m^2) - x(2a+2bm) = 0</math> ✓<br/> <math>x = 0</math> or <math>x = \frac{2a+2bm}{1+m^2}</math> ✓<br/> <math>x</math>-coord of P is <math>\frac{2(a+bm)}{1+m^2}</math> (2)</p> <p>(ii) <math>y</math>-coord. of P is <math>\frac{2m(a+bm)}{1+m^2}</math> ✓<br/> <math>M: \left( \frac{a+bm}{1+m^2}, \frac{m(a+bm)}{1+m^2} \right)</math> ✓ (2)</p> <p>(iii) <math>x = \frac{a+bm}{1+m^2}</math> <math>y = mx</math><br/> <math>\therefore m = \frac{y}{x}</math><br/> <math>x(1+m^2) = a+bm</math><br/> <math>x\left(1 + \frac{y^2}{x^2}\right) = a + b\left(\frac{y}{x}\right)</math> ✓<br/> <math>x^2 + y^2 = ax + by</math> ✓<br/> <math>\left(x^2 - ax + \frac{a^2}{4}\right) + \left(y^2 - by + \frac{b^2}{4}\right) = \frac{a^2}{4} + \frac{b^2}{4}</math><br/> <math>\left(x - \frac{a}{2}\right)^2 + \left(y - \frac{b}{2}\right)^2 = \frac{a^2+b^2}{4}</math> ✓<br/>         Locus is a circle, centre <math>\left(\frac{a}{2}, \frac{b}{2}\right)</math><br/>         radius <math>\frac{\sqrt{a^2+b^2}}{2}</math> ✓ (4)</p> | <p>1 mark for part of the solution<br/>         eg <math>\frac{5\pi}{3} \leq \theta \leq \frac{7\pi}{3}</math></p> <p>or equivalent.</p> |

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Comments                            |
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| <p><u>QUESTION 7 (cont.).</u></p> <p>(c). </p> <p>(i) <math>\angle QPB = \angle BPR</math> because angles at the <u>circumference</u> standing on equal arcs are equal. (1)</p> <p>(ii) Let <math>\angle QPB = \angle BPR = \alpha</math> <sup>at circumference</sup><br/> <math>\angle BQR = \angle BPR = \alpha</math> (angles on same arc) ✓<br/> <math>\angle AQB = \angle ABP = \beta</math> (angles at circumference on equal arcs) ✓<br/> <math>\angle QDC = \angle QBD + \angle DQB</math> (ext. angle <math>\triangle QDB</math>) ✓<br/> <math>= \beta + \alpha</math><br/> <math>\angle QCD = \angle CPB + \angle CBP</math> (ext. angle <math>\triangle BCP</math>) ✓<br/> <math>= \alpha + \beta</math><br/> <math>\therefore \angle QDC = \angle QCD</math><br/> <math>\therefore QC = QD</math>. (4)</p> | <p>Must mention "circumference"</p> |



| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Comments                                                                                                                                    |
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| <p><u>QUESTION 8</u></p> <p>a) <math>x = V \cos \theta t</math> , <math>y = V \sin \theta t - \frac{1}{2} g t^2</math></p> <p>(i) Subst. <math>x = d</math>, <math>y = h</math> and eliminate <math>t</math>.</p> $d = V \cos \theta t$ $h = V \sin \theta t - \frac{1}{2} g t^2$ $t = \frac{d}{V \cos \theta}$ $h = V \sin \theta \cdot \frac{d}{V \cos \theta} - \frac{g}{2} \cdot \frac{d^2}{V^2 \cos^2 \theta}$ $h = d \tan \theta - \frac{g d^2}{2 V^2 \cos^2 \theta}$ $\frac{g d^2}{2 V^2 \cos^2 \theta} = d \tan \theta - h$ $V^2 = \frac{g d^2}{2 \cos^2 \theta (d \tan \theta - h)}$ $= \frac{g d^2 \sec^2 \theta}{2 (d \tan \theta - h)} \quad (2)$ <p>(ii) <math>y = V \sin \theta t - \frac{1}{2} g t^2</math></p> $= 0 \text{ when } t = \frac{V \sin \theta}{g}$ <p>Max. ht. = <math>V \sin \theta \times \frac{V \sin \theta}{g} - \frac{g}{2} \times \frac{V^2 \sin^2 \theta}{g^2}</math></p> $= \frac{V^2 \sin^2 \theta}{2g}$ $= \frac{g d^2 \sec^2 \theta}{2 (d \tan \theta - h)} \times \frac{\sin^2 \theta}{2g}$ $= \frac{d^2 \tan^2 \theta}{4 (d \tan \theta - h)} \quad \checkmark \text{ since } \sec^2 \theta \sin^2 \theta = \frac{1}{\cos^2 \theta} \sin^2 \theta = \tan^2 \theta$ <p>(iii) <math>h = \frac{d^2 \tan^2 \theta}{4 (d \tan \theta - h)}</math></p> $d^2 \tan^2 \theta = 4h (d \tan \theta - h)$ $d^2 \tan^2 \theta - 4dh \tan \theta + 4h^2 = 0$ $(d \tan \theta - 2h)^2 = 0$ $d \tan \theta - 2h = 0$ $\therefore \tan \theta = \frac{2h}{d} \quad \checkmark \quad (3)$ | <p>OR Subst. <math>\tan \theta = \frac{2h}{d}</math> into (iii) and show it equals 'h'.<br/>Need description conclusion for third mark.</p> |

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Comments                                                                                                                          |
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| <p><u>QUESTION 8 (cont.)</u></p> <p>(b)(i) Prove <math>(\sqrt{3}-1)^n = p_n + q_n \sqrt{3}</math>; <math>p_n, q_n</math> integers</p> <p>When <math>n=1</math>, LHS = <math>(\sqrt{3}-1)^1 = -1 + \sqrt{3}</math>.</p> <p><math>p_1 = -1</math>, <math>q_1 = 1</math> are unique <math>\therefore</math> true for <math>n=1</math></p> <p>Assume it is true for <math>n=k</math>,<br/>i.e. assume <math>(\sqrt{3}-1)^k = p_k + q_k \sqrt{3}</math> (<math>p_k, q_k</math> unique integers)</p> <p>When <math>n=k+1</math>,<br/>LHS = <math>(\sqrt{3}-1)^{k+1}</math></p> $= (\sqrt{3}-1)(\sqrt{3}-1)^k$ $= (\sqrt{3}-1)(p_k + q_k \sqrt{3}) \quad \checkmark \text{ by assumption}$ $= p_k \sqrt{3} + 3q_k - p_k - q_k \sqrt{3}$ $= (3q_k - p_k) + (p_k - q_k) \sqrt{3}$ $= p_{k+1} + q_{k+1} \sqrt{3} \quad \checkmark$ <p>where <math>p_{k+1} = 3q_k - p_k</math>, <math>q_{k+1} = p_k - q_k</math></p> <p><math>\therefore</math> If it is true for <math>n=k</math>, it is true for <math>n=k+1</math></p> <p>Since it is true for <math>n=1</math>, it is true for <math>n=2, 3, \dots</math> (5)</p> <p>(ii) <math>p_n^2 - 3q_n^2 = (3q_{n-1} - p_{n-1})^2 - 3(p_{n-1} - q_{n-1})^2</math></p> $= 9q_{n-1}^2 - 6q_{n-1}p_{n-1} + p_{n-1}^2 - 3p_{n-1}^2 + 6p_{n-1}q_{n-1} - 3q_{n-1}^2$ $= 6q_{n-1}^2 - 2p_{n-1}^2$ $= -2(p_{n-1}^2 - 3q_{n-1}^2) \quad \checkmark$ $= (-2)(-2)(p_{n-2}^2 - 3q_{n-2}^2)$ $= (-2)(-2) \dots (-2)(p_1^2 - 3q_1^2)$ $= (-2)(-2) \dots (-2)(1-3)$ $= (-2)^n \quad \checkmark \quad (2)$ | <p>Note: Mark for conclusion not awarded if second last mark not awarded</p> <p>OR<br/>Prove by induction (but only 2 marks).</p> |